

TimeTensors Experiment 6: Central versus Per-user Forecasting

1 Theoretical overview and goal

A central model estimates one mapping from all users and can transfer common dynamics; a per-user model specializes but sees less data. Window-weighted MSE can favor users with more accessible windows, so the experiment also uses the equal-user mean and the worst-user tail

$$\text{W10}(\ell) = \frac{1}{\lceil 0.1U \rceil} \sum_{u \in \text{Top}_{\lceil 0.1U \rceil}} \bar{\ell}_u.$$

Experimental goal. Measure the benefit and fairness cost of cross-user learning, and compare specialization with shared forecasting for trained and pretrained backbones.

2 Implementation details

PatchTST is trained centrally or independently per user. Chronos central inference enables cross-learning across the batch; per-user inference disables it. All variants use identical temporal splits and evaluation queries. The combined table reports ordinary MSE, equal-user MSE, and W10.

This is a later-stage sweep after the data and optimization choices are fixed. The study profile spans six datasets, three settings, three seeds, and both PatchTST and Chronos. PatchTST uses the common 10^{-5} learning rate, batch size 256, 10,000 optimizer steps, and 1,000-step validation. The test profile runs one PatchTST Electricity case.

Launcher: `07_central_per_user.slurm`

Per-user training creates multiple fitted models and must retain user identity through evaluation. Central and per-user results intentionally share one table because the training scope is the independent variable.

3 Interpretation

A lower window-weighted MSE with worse W10 indicates that cross-user learning benefits the average while harming the tail. Per-user gains may reflect specialization but should be interpreted with their greater model count and lower data per fit. Chronos cross-learning is an inference-mode comparison, whereas PatchTST is a training-scope comparison.